DESIGN MILESTONES

DERAKHT

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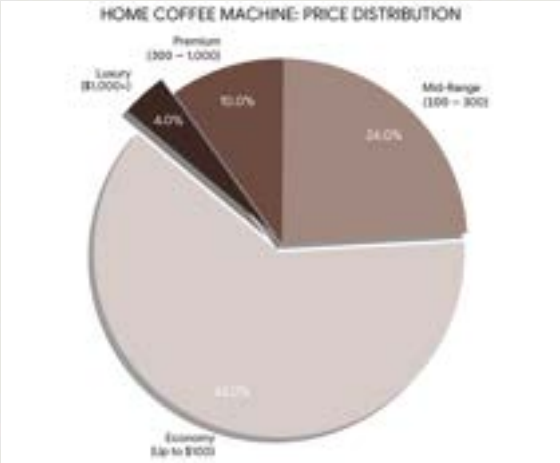
IDE402

1 RESEARCH

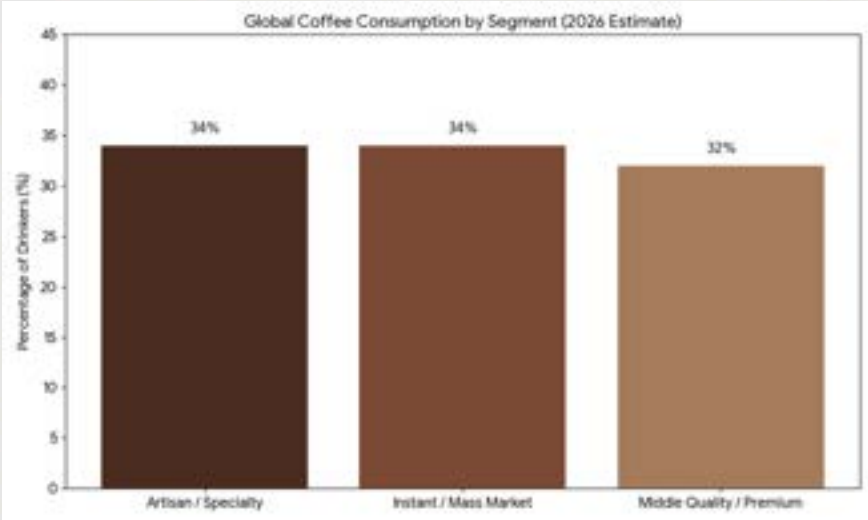
for the research, I used AI to find ritualistic hot beverages from different cultures throughout the world. Among all the different beverages, coffee stood out because:

- It is a universal "umbrella drink" that transcends borders. By exploring diverse brewing rituals and cultural traditions, this project can draw inspiration from the global intersection of coffee and community.
- Global coffee culture and experience remains deeply fragmented across brewing rituals, technical methods, and bean quality.

Upon further research, I made the AI scan through Youtube videos and the web to find me all the coffee brewing methods. Seeing the advantages and disadvantages of different coffee brewing methods directed my design thinking.



- **The Ritual Gap:** While enthusiasts use "lab-grade" artisan gear, everyday users are excluded from the coffee ritual, frequently relying on low-quality alternatives like soluble coffee.
- **Market Absence:** No deliberate, high-quality brewers exist that cater specifically to the non-enthusiast.



- **Global Scale:** 1.25 billion people consume coffee daily.
- **The Functional Trap:** 34% of consumers rely on mass-market/instant coffee, viewing it only as a functional stimulant.
- No significant brewing methods are invented in 21st century.
- Make the user directly a part of the coffee making process. Replace the passive button pushing user interface with active and involved user interface

Coffee Brewing Pain Points

- French Press: Difficult to clean.
- Pour Over: Inconsistent results.
- Clever Dripper: Challenging maintenance.
- AeroPress: Limited serving capacity.
- Moka Pot: Lack of brewing control.
- Jezve / Ibrik: High risk of over-extraction.
- Drip Machine: Large counter footprint.
- Espresso: Expensive equipment costs.
- Capsules: Significant environmental waste.

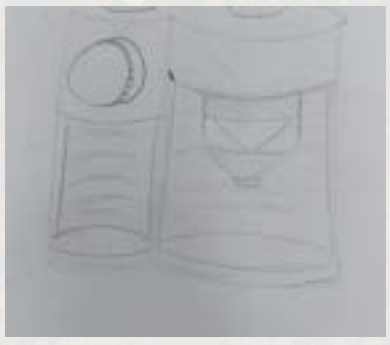
Interview Insights

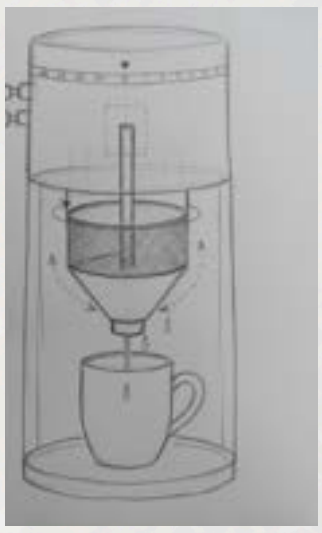
The interview was made with one of the founders of Ozu Brew Academy.
The answers were noted on a notepad and uploaded to AI to analyse.

Key Points:

- Every brewing method has different advantages. Espresso lets out the flavors of the oils more while filter coffee opens up the floral and milder flavours.
- ın artisan coffee brewing, the more we have the contorl of every element, the better experience and brew becomes.
- There are a type of coffee consumers called “Tchibo Kahvecisi” who enjoy coffee but haven’t stepped to being a coffee enthusiast or a hobbyist yet

2 SKETCHES





3 DECISION AND PROTORYPE

After the sketching phase using mostly Vizcom, I decided to use “Centrifugical extraction” method which can make espresso to filter coffee and everything in between.
I also wanted the coffee machine to have a grinder.
The coffee machine had to offer the user full access to all the phases of the brewing process.

- First, the spinning coffee basket assembly was designed.
- This is a portafilter type part that connects to the coffee machine and spins to extract the coffee, funneling the it to the mug from the bottom.
- The gear tooth on the basket are connected to the machine’s gear system in order to spin

DESIGN ITERATIONS OF THE PROTOTYPE

- The challenge of working with different iterations is to find the most suitable from hundreds of alternatives
- The initial idea for this tree- type design emerged from deciding on the components first. The components are the definitive aspect of this form.

4 FINAL DESIGN

THE DERAKHT COFFEE MAKER

- The DERAKHT Coffee Maker is a Centrifugical Extraction coffee machine which has the capability of making espresso, filter coffee and everything in between.
- It has 3 main stations:

1. Grinding: The left cylinder houses the coffee beans and has a ring dial to adjust the grind size. It has a button to start the grinding
2. Water: the middle cylinder houses the water and has a ring dial to choose the right temperature. It has a start button on the bottom
3. Centrifuge: the right cylinder houses the water pump and the brushless electric motor that spins the basket on the portafilter.

DERAKHT

BRANDING

- “Derakht” means tree in Persian
- Lone tree or “Tak Derakht” symbolizes solitude and nobility in the Persian Literature
- Just like a lone tree, DERAKHT coffee maker can brew all the coffee types just by itself.
- The coffee enthusiast needs “lonely” this coffee machine

- Centrifuge and extraction cylinder

- water heating cylinder

- Coffee bean storage and grinder cylinder